Ain Shams University, Faculty of Engineering

Machine Learning Milestone 1

To Dr. Mahmoud Khalil & Eng. Ahmed Nezar

Group 2 Section 3

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1.0 Milestone 1 Objective

1. Load and verify H2.csv; complete data understanding and EDA, including missing values, duplicates, variable types, unusual values, distributions, and suitable plots.

2. Define the prediction point, task-specific feature choices, and leakage risks for regression, classification, and clustering. Document a reproducible train/validation/test or cross-validation strategy.

3. Implement reusable preprocessing pipelines, including justified missing-value handling, categorical encoding, scaling, and any feature engineering. Fit data-dependent supervised preprocessing on training data only.

4. Run initial untuned experiments with at least one regression model, one classification model, and one clustering algorithm. Report appropriate preliminary metrics and a brief interpretation; keep the final supervised test set untouched for final evaluation.

5. Submit a progress report containing completed work, initial results, a preliminary leakage audit, member contributions, and the plan for the remaining models, tuning, PCA/LDA, and final application.

2.0 EDA

2.1 Features

2.1.1 Descriptive Statistics

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
IsCanceled	79330				0.417269633	0.493111297	0	0	0	1	1
LeadTime	79330				109.7357242	110.9485258	0	23	74	163	629
ArrivalDateYear	79330				2016.174285	0.699181325	2015	2016	2016	2017	2017
ArrivalDateMonth	79330	12	August	8983							
ArrivalDateWeekNumber	79330				27.17744863	13.39852289	1	17	27	38	53
ArrivalDateDayOfMonth	79330				15.78662549	8.728450546	1	8	16	23	31
StaysInWeekendNights	79330				0.795184672	0.885026383	0	0	1	2	16
StaysInWeekNights	79330				2.182957267	1.456416186	0	1	2	3	41
Adults	79330				1.850976932	0.509292308	0	2	2	2	4
Children	79326				0.09136979	0.37217687	0	0	0	0	3
Babies	79330				0.004941384	0.084323341	0	0	0	0	10
Meal	79330	4	BB	62305							
Country	79306	166	PRT	30960							
MarketSegment	79330	8	Online TA	38748							
DistributionChannel	79330	5	TA/TO	68945							
IsRepeatedGuest	79330				0.025614522	0.15798333	0	0	0	0	1
PreviousCancellations	79330				0.079742846	0.415472214	0	0	0	0	21
PreviousBookingsNotCanceled	79330				0.132371108	1.693410601	0	0	0	0	72
ReservedRoomType	79330	8	A	62595							
AssignedRoomType	79330	9	A	57007							
BookingChanges	79330				0.187369217	0.608620314	0	0	0	0	21
DepositType	79330	3	No Deposit	66442							
Agent	79330	224	9	31955							
Company	79330	208	NULL	75641							
DaysInWaitingList	79330				3.226774234	20.87089	0	0	0	0	391
CustomerType	79330	4	Transient	59404							
ADR	79330				105.3044654	43.60295384	0	79.2	99.9	126	5400
RequiredCarParkingSpaces	79330				0.02436657	0.154919473	0	0	0	0	3
TotalOfSpecialRequests	79330				0.546917938	0.780776222	0	0	0	1	5
ReservationStatus	79330	3	Check-Out	46228							
ReservationStatusDate	79330	864	10/21/2015	1416							

2.1.2 Checking for duplicates and missing values

	dtype	non_null	missing_count	missing_percent	unique_values
IsCanceled	int64	79330	0	0	2
LeadTime	int64	79330	0	0	453
ArrivalDateYear	int64	79330	0	0	3
ArrivalDateMonth	object	79330	0	0	12
ArrivalDateWeekNumber	int64	79330	0	0	53
ArrivalDateDayOfMonth	int64	79330	0	0	31
StaysInWeekendNights	int64	79330	0	0	14
StaysInWeekNights	int64	79330	0	0	29
Adults	int64	79330	0	0	5
Children	float64	79326	4	0.01	4
Babies	int64	79330	0	0	5
Meal	object	79330	0	0	4
Country	object	79306	24	0.03	166
MarketSegment	object	79330	0	0	8
DistributionChannel	object	79330	0	0	5
IsRepeatedGuest	int64	79330	0	0	2
PreviousCancellations	int64	79330	0	0	10
PreviousBookingsNotCanceled	int64	79330	0	0	73
ReservedRoomType	object	79330	0	0	8
AssignedRoomType	object	79330	0	0	9
BookingChanges	int64	79330	0	0	21
DepositType	object	79330	0	0	3
Agent	object	79330	0	0	224
Company	object	79330	0	0	208
DaysInWaitingList	int64	79330	0	0	115
CustomerType	object	79330	0	0	4
ADR	float64	79330	0	0	5405
RequiredCarParkingSpaces	int64	79330	0	0	4
TotalOfSpecialRequests	int64	79330	0	0	6
ReservationStatus	object	79330	0	0	3
ReservationStatusDate	object	79330	0	0	864

Number of duplicate rows: 25902

IsCanceled	With duplicates	Without duplicates
0	58.273037%	69.961443%
1	41.726963%	30.038557%

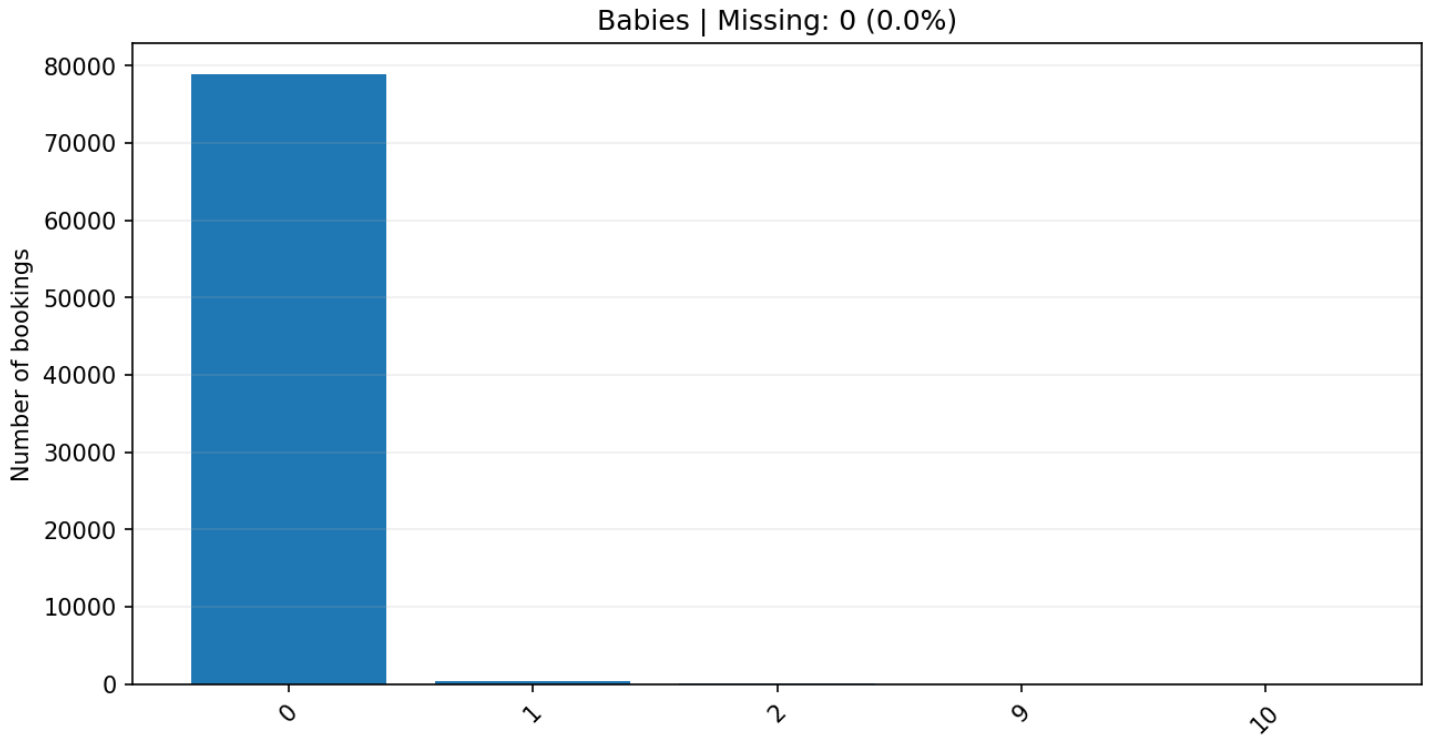
We decided to not remove the duplicates, as they are not real duplicated data, but multiple people with the same booking details, which is normal for this dataset to have duplicates.

2.2 Visualization

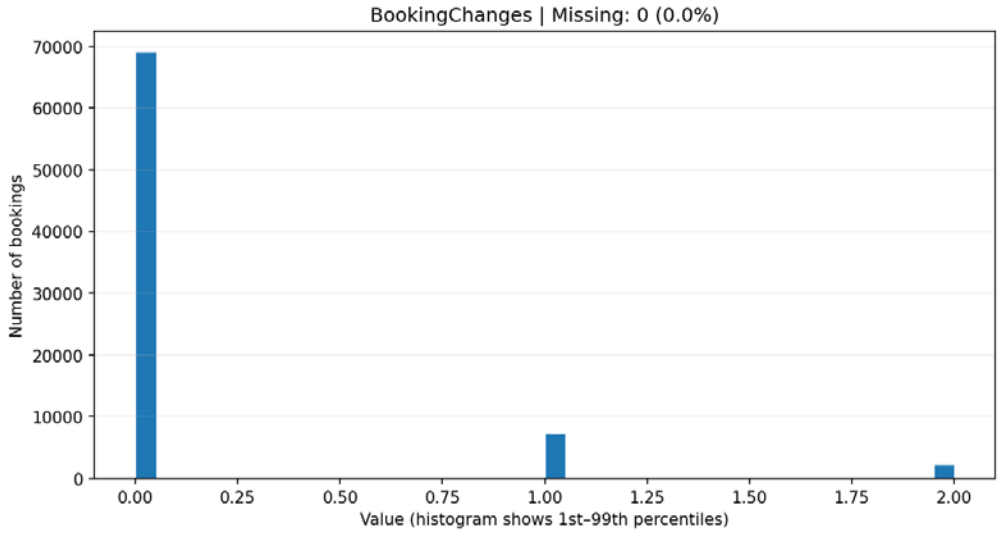
2.2.1 Removing unbalanced features

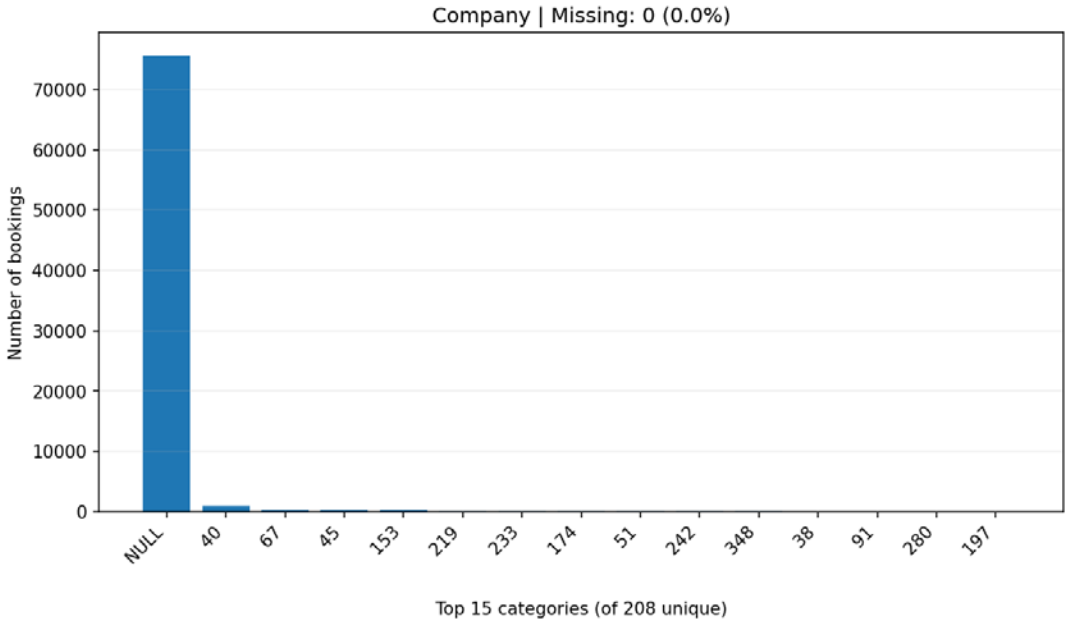
First, we created plots showing the distribution (histograms/barcharts) for all features, in order to examine them, to remove heavily unbalanced features and data that doesn't make any sense.

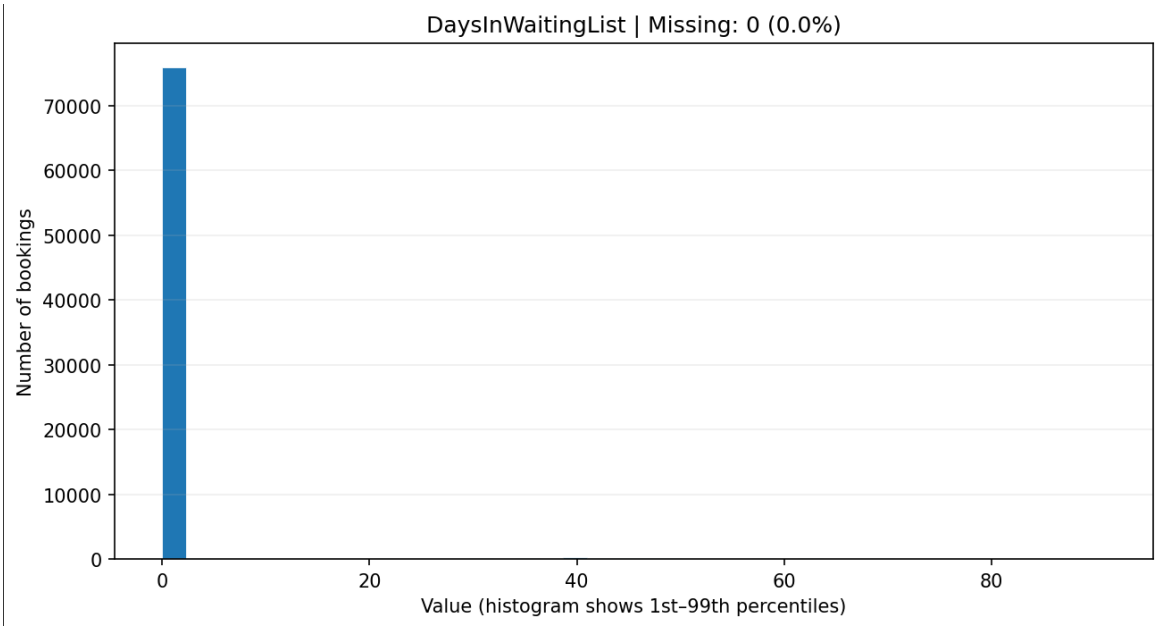
For example, for the “Babies” feature, almost all the data has 0 as its value, so this feature is clearly unneeded and holds no meaning.

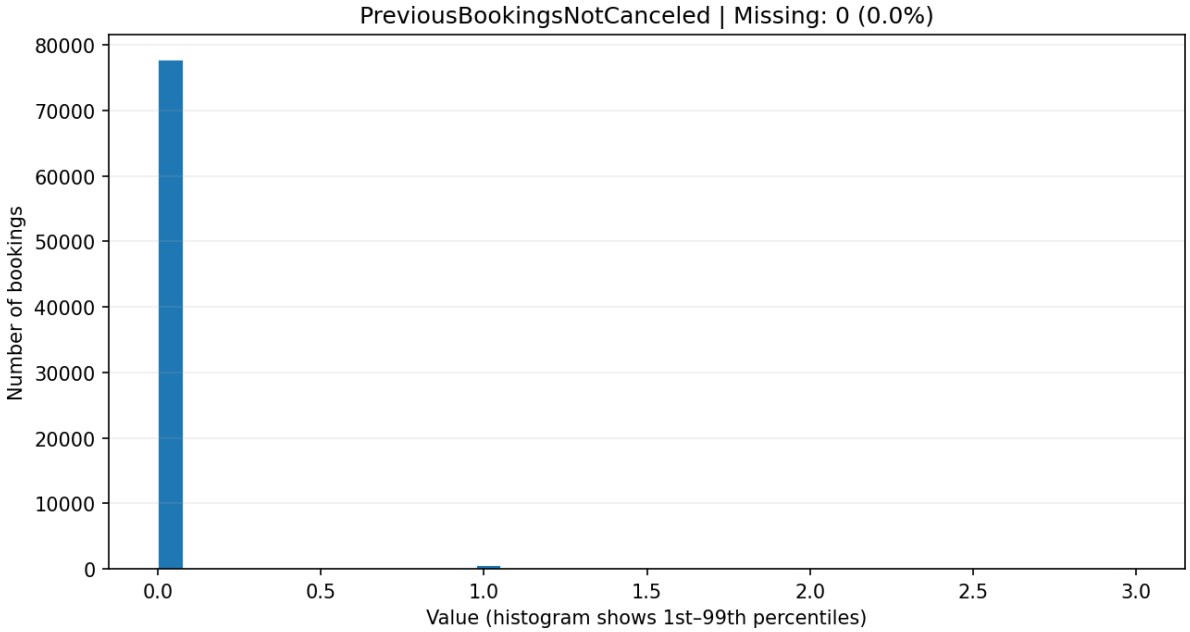


The same can be said for the following features: BookingChanges, Company, DaysInWaitingList, PreviousBookingsNotCanceled


BookingChanges | Missing: 0 (0.0%)
Number of bookings
70000
60000
50000
40000
30000
20000
10000
0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Value (histogram shows 1st–99th percentiles)

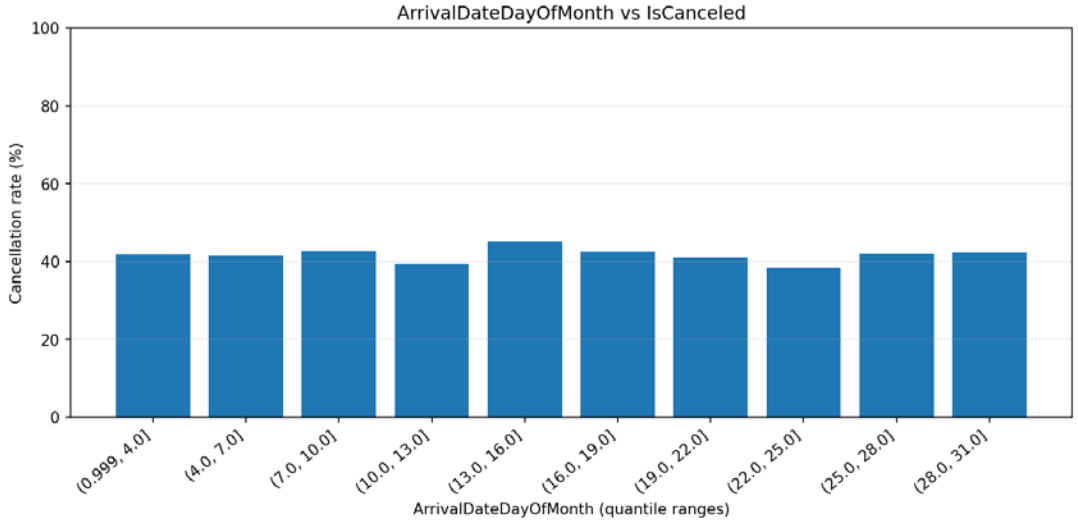

Company | Missing: 0 (0.0%)
Number of bookings
70000
60000
50000
40000
30000
20000
10000
0
NULL
40
67
45
153
219
233
174
51
242
348
38
91
280
197
Top 15 categories (of 208 unique)

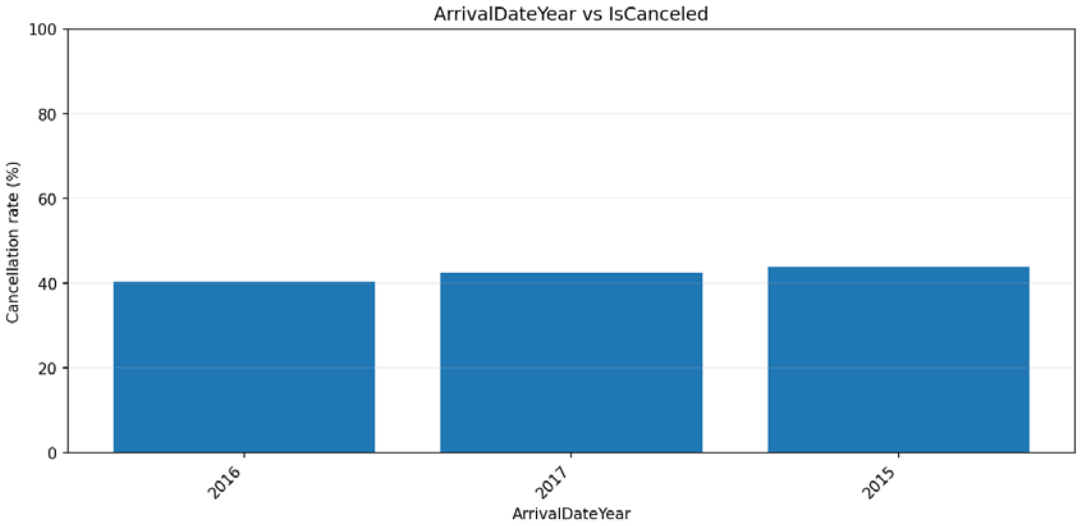

DaysInWaitingList | Missing: 0 (0.0%)
Number of bookings
70000
60000
50000
40000
30000
20000
10000
0
0
20
40
60
80
Value (histogram shows 1st–99th percentiles)


PreviousBookingsNotCanceled | Missing: 0 (0.0%)
Number of bookings
80000
70000
60000
50000
40000
30000
20000
10000
0
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Value (histogram shows 1st–99th percentiles)

2.2.2 Data with “random noise”

Data with “random noise” seem to be constant and have 0 correlation to anything other than to itself (this is further visualized in the correlation matrix below). Features like that are random noise are: ArrivalDateDayOfMonth, ArrivalDateYear

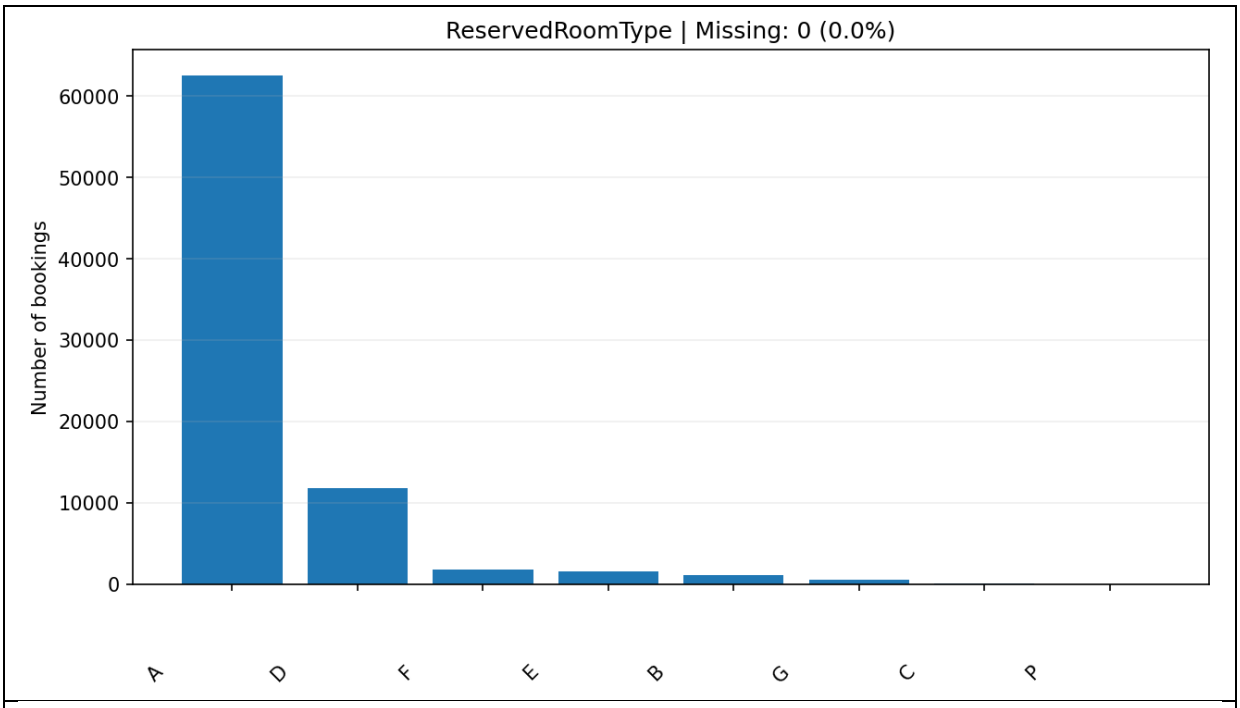


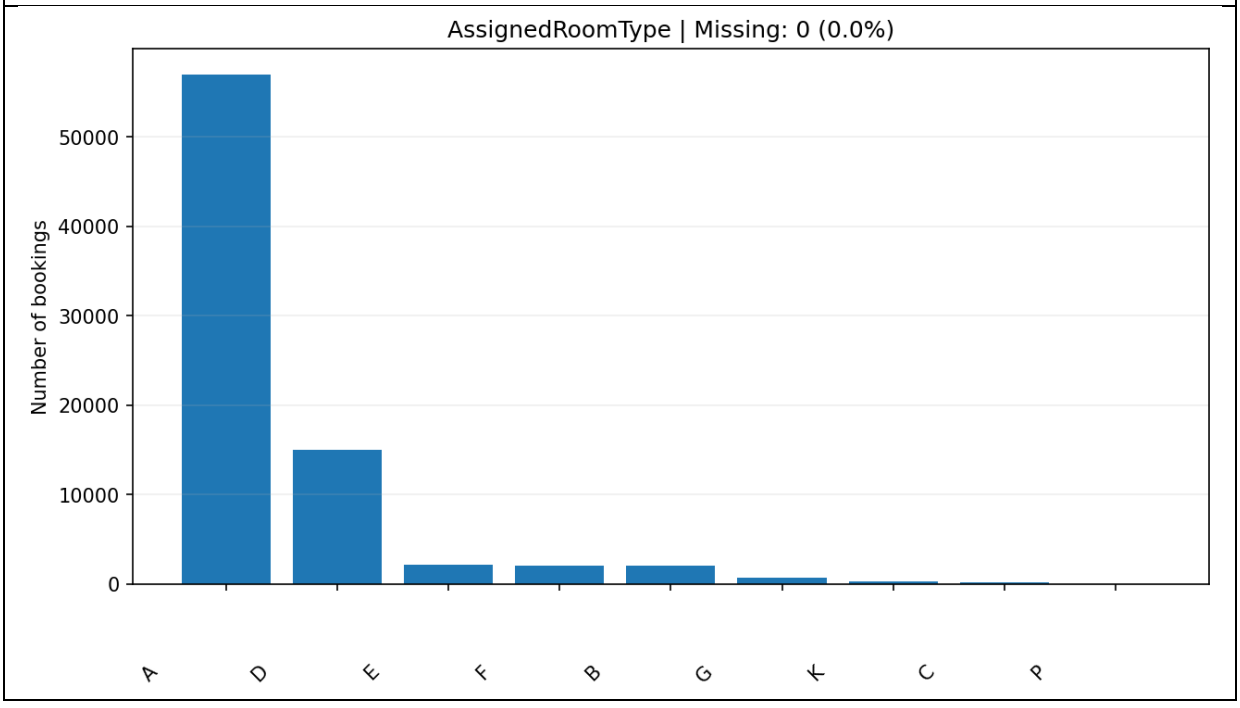


2.2.3 Identical Feature Columns

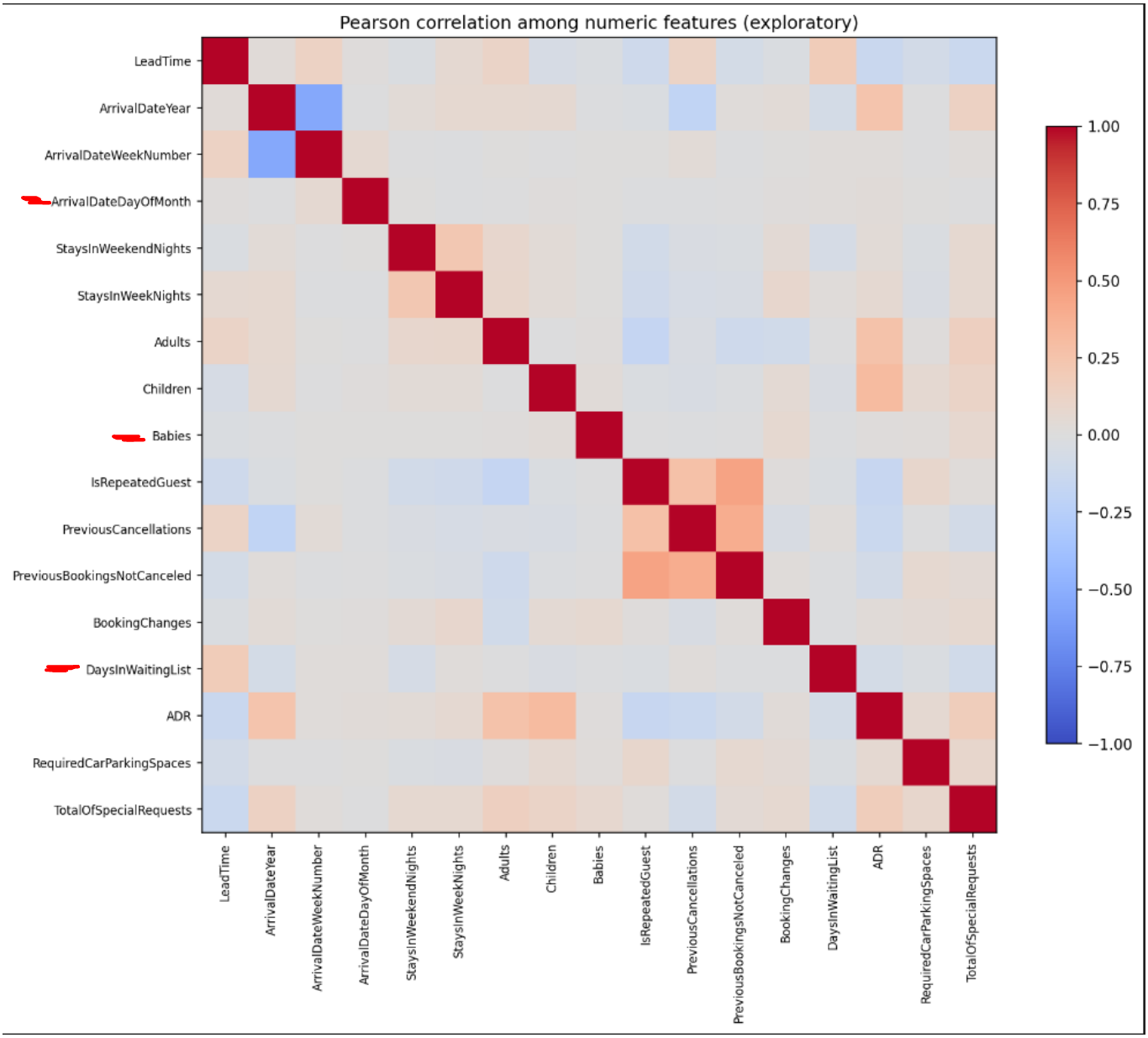
PCA would catch something like this, useful to visualize

(ReserveredRoomType & AssignedRoomType)





2.2.4 Correlation



(Marked in red are some examples of worthless data we dropped, with near-zero correlation to any other features)

2.2.5 Some Notes

Our 3 columns, “Adults” “Children” and “Babies”, we removed all rows where the sum of these 3 columns are 0. In english, we removed all hotel bookings where there are zero listed guests. (167 rows were removed)

We considered adding the 2 columns “StaysInWeekendNights” and “StayInWeekNights” to 1 column for total number of booked days, but decided to let the model itself figure out the pattern.

In the train-validate split, we will make sure that duplicates are isolated in one of the sections so the model doesn't cheat in the validating phase.

3.0 Model Training

3.1 Cardinality & One-Hot Encoding

One-Hot encoding is where we expand unqiue values into different columns. For example a “Season” feature would have 4 unique values, if we one-hot-encode this “Season” column, it will expand into 4 new columns (Winter, Summer, Spring, Fall) which would be a boolean true/false value where of course only one of them will be true and all the others false. This is EXTREMELY detrimental if there are many unique values which will explode in dimension which makes model training harder.

We chose our cutoff of number of unique values to be 12 or less. Features with 12 or less will be called “low cardinality features”

The features we one-hot encoded:

```
low_cardinality_features = [
    "AssignedRoomType",
    "MarketSegment",
    "DistributionChannel",
    "Meal",
    "CustomerType",
    "DepositType",
    "ArrivalDateMonth"
]
```

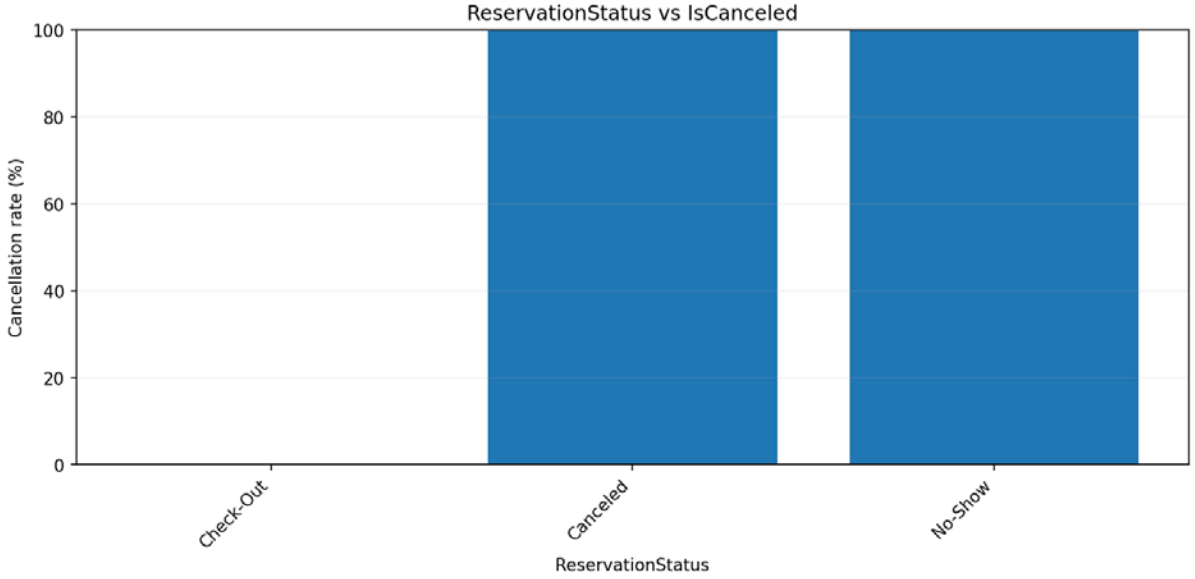
High-cardinality features with many unique categorical values, we will group these values by frequency, values with very low frequencies will be grouped into 1 value. Our cutoff was 1% frequency. All unique values with 1% or less frequency will have an “Other” value

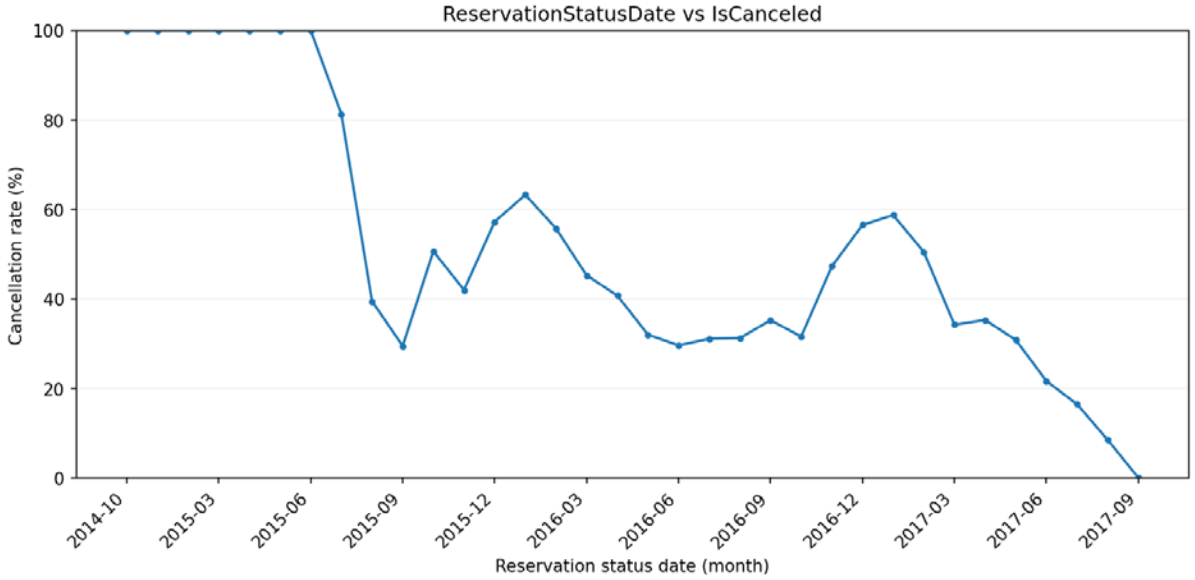
Low-cardinalities are split into more COLUMNS

High-cardinalities are grouped into less UNIQUE VALUES

3.2 Data Leakage

Data leakage is when the “target” feature for our model is hidden within other features which allows the model to technically cheat the answer rather than finding the underlying pattern. For our classification where “IsCanceled” is our target, “ReservationStatus” where it literally says if the reservation cancelled, and “ReservationStatusDate” where it says when the reservation status was last updated.





3.3 Linear Regression

We trained simple classification model with/without PCA/LDA and these were the results

PCA took the number of features/dimensions down from 91 to 35

```
=== Original Classification Model ===
Accuracy: 0.8128
ROC-AUC:  0.9021

=== PCA Classification Model ===
Accuracy: 0.8103
ROC-AUC:  0.8952

=== LDA Classification Model ===
Accuracy: 0.8106
ROC-AUC:  0.8998

=== Original Classification Model (Non-Duplicated) ===
Accuracy: 0.7759
ROC-AUC:  0.8254

=== PCA Classification Model (Non-Duplicated) ===
Accuracy: 0.7723
ROC-AUC:  0.8144

=== LDA Classification Model (Non-Duplicated) ===
Accuracy: 0.7739
ROC-AUC:  0.8211
```

3.4 Linear Regression

A simple regression model for “ADR” with root mean square error function

PCA took the number of features/dimensions down from 91 to 35

```
=== Original Regression Model ===
RMSE:    47.3918
R2 Score: 0.2964

=== PCA Regression Model ===
RMSE:    49.5601
R2 Score: 0.2305

=== Original Regression Model (Non-Duplicated) ===
RMSE:    56.8151
R2 Score: 0.2654

=== PCA Regression Model (Non-Duplicated) ===
RMSE:    58.7293
R2 Score: 0.2150
```

3.5 Clustering (K-Means)

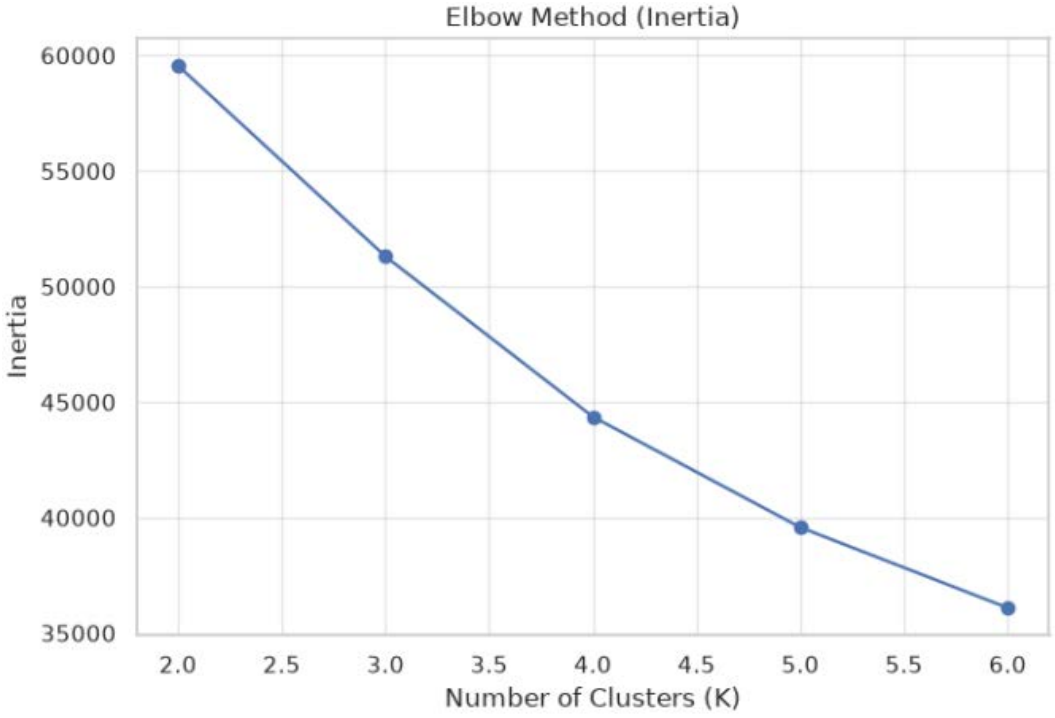
We decided to cluster about these features With these as the “leaking” features

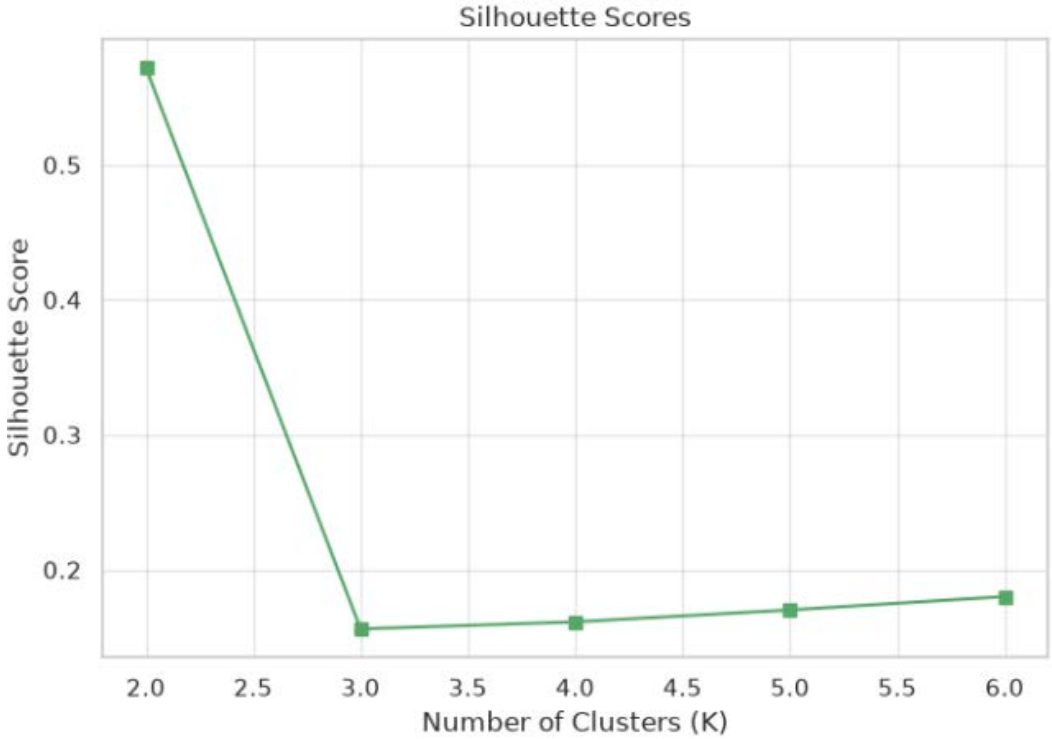
```
numeric_features = [
    'LeadTime',
    'StaysInWeekendNights',
    'StaysInWeekNights',
    'TotalGuests',
    'TotalOfSpecialRequests',
    'RequiredCarParkingSpaces'
]

categorical_features = [
    'MarketSegment',
    'CustomerType',
    'ReservedRoomType',
    'Meal',
    'DistributionChannel',
    'IsRepeatedGuest'
]
```

```
leakage_cols = [
    'IsCanceled',
'ReservationStatus',
'ReservationStatusDate',
    'ArrivalDateYear',
'ArrivalDateMonth',
'ArrivalDateWeekNumber',
'ArrivalDateDayOfMonth'
]
```

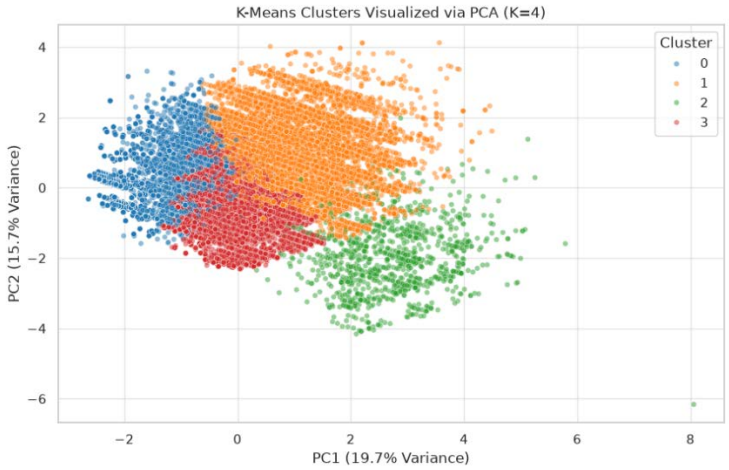
For K-Means, we need to decide what k is equal to (number of clusters).

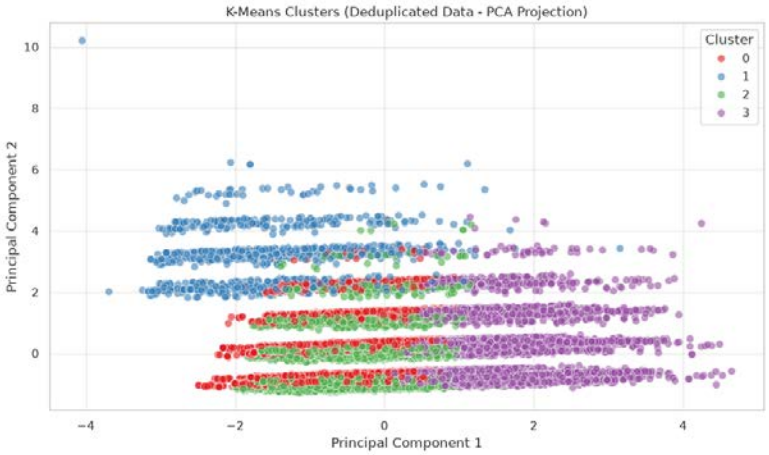




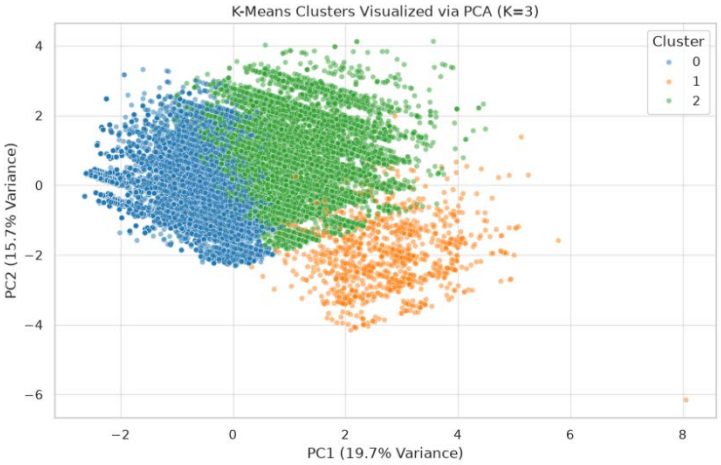
The clear candidates for K is k=3 from the silhouette scores, and k=4 from elbow method, so we tried to visualize both of these for both duplicated and deduplicated data.

For k=4, the model becomes largely colorblind, even more so in the deduplicated version. K=3 is likely better





k=3, the clusters look way more reasonable than whatever happened with k=4 above^^





4.0 Github Repository

https://github.com/Hassan-El-Hakim/ML-Project